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The Aletheion Continuity Protocol: A Multi-Layered Framework for Seamless Transition from Files 146-200

Ecological Data Sovereignty and Research Prioritization for Tier 2

The successful progression of the Aletheion codebase into Files 151-175 is contingent upon establishing a foundation of verifiable ecological data and a clear procedural path for governance compliance. The project plan correctly identifies these areas as critical research gaps that must be resolved before code generation can commence . This section outlines the specific data requirements, their strategic importance, and a prioritized approach to research and modeling that aligns with the project's core mandates of ecological accuracy and treaty adherence. The analysis prioritizes the user's directive for ecological accuracy in cyber-aquatic system integrations and VM-node establishment, using these as foundational models before scaling to broader agricultural applications.

The primary obstacle blocking the generation of Tier 2 files is the absence of domain-specific data required for accurate simulation and implementation . These gaps span multiple domains, including soil science, atmospheric water harvesting, crop viability, and traditional ecological knowledge. Without this data, any code written would be speculative and non-compliant. For instance, the `Sonoran Crop Viability` module (Files 161-170) requires detailed information on heat-tolerant crops capable of surviving Phoenix's extreme temperatures, which are noted to reach 120°F or higher . Similarly, the design of infrastructure like `Monsoon Capture Systems` depends on validated rainfall data, such as the 2.71 inches recorded in a September 2025 event, to ensure engineering specifications are robust and effective . The most critical gap identified is the need for formal Free, Prior, and Informed Consent (FPIC) consultation regarding the integration of Piipaash Traditional Farming knowledge, which is not merely a data collection task but a procedural requirement tied to the Akimel O'odham/Piipaash Treaties and governance frameworks [28] [29].

To address these challenges, a phased research strategy is recommended. The immediate action, as per the existing plan, is to initiate parallel tasks for the Ag, Env, Energy, and

Treaty teams to resolve the identified gaps . However, to accelerate progress and validate architectural concepts simultaneously, it is strategically sound to begin by modeling a small-scale cyber-aquatic system. This exercise serves as a testbed for the core principles of ecological modeling and virtual machine (VM) node establishment, which are explicitly prioritized by the user. Such a model could involve simulating a controlled aquatic environment, like a bio-dome, where key variables such as temperature, nutrient levels, and oxygen saturation are regulated. By applying established algorithms for state estimation, such as the Kalman filter and its extensions for nonlinear systems, a predictive model of this closed-loop system's dynamics can be developed [1]. This model, when implemented within a VM-node framework, would allow for testing the responsiveness, scalability, and stability of the underlying control logic before application to more complex and larger-scale systems like desert agriculture. This foundational work provides a template and a proof-of-concept for the entire agricultural layer, ensuring that the subsequent modeling of Sonoran Desert ecosystems is built upon a validated and understood methodology.

The following table details the critical research gaps for Tier 2 (Files 151-175) and their direct implications for code generation and compliance. It is imperative that each item in this table is addressed through targeted research before its corresponding file is generated.

File Range	Domain	Research Requirement	Status	Owner	Implication for Code Generation
151-160	Soil & Water	Maricopa County Soil Composition (pH, salinity, nutrients)	⚠️ Gap	Ag Team	Required for `SoilMoistureRegulator` and `NutrientDeliveryLogic` in crop modules. Lack of data makes these modules speculative.
151-160	Soil & Water	MOF Water Harvesting Specs (MIT 2025 field tests vs. Phoenix humidity)	⚠️ Gap	Env Team	Essential for calculating yield and feasibility of atmospheric water capture systems in `AtmosphericWaterHarvester`.
151-160	Soil & Water	Monsoon Capture Systems Engineering Specs (Sept 2025 rainfall data)	⚠️ Gap	Env Team	Critical for designing physical infrastructure for `FlashFloodAvoidance` and `RainwaterCollection`.
161-170	Crop Systems	Sonoran Crop Viability (Heat tolerance >120°F)	⚠️ Gap	Ag Team	Core input for `CropSelectionEngine` and `DesertAgriculturePlanner`. Determines the very possibility of the agricultural layer.
161-170	Crop Systems	Piipaash Traditional Farming Knowledge	⚠️ Critical	Treaty Office	Requires formal FPIC consultation. Integration is mandated by Akimel O'odham/Piipaash Treaties and G-Compliance.
171-175	Distribution	Phoenix Food Desert Maps (USDA 2025 access data)	✅ Ready	Logistics	Needed for optimizing distribution routes and equity scoring in `FoodDistributionNetwork`.
171-175	Distribution	Cold Chain Logistics Energy Requirements (>120°F ambient)	⚠️ Gap	Logistics	Necessary for designing sustainable refrigeration solutions and ensuring power grid resilience.

This structured approach ensures that all development is grounded in reality, mitigating risks associated with unproven assumptions and guaranteeing that Aletheion remains compliant with its environmental, social, and legal obligations. The focus on a foundational cyber-aquatic model allows for iterative refinement of the core simulation engine while other teams gather the necessary data for the broader agricultural systems, creating a parallel and efficient workflow.

Architectural Blueprint for a Multi-Language Codebase

A strict and well-defined language policy is essential for maintaining code quality, performance, and long-term maintainability. The directive to exclude Python entirely and mandate a specific division of labor among Rust, C++, and Lua is a cornerstone of Aletheion's architectural integrity . This section provides a detailed blueprint for implementing this multi-language architecture, defining the precise role of each language and outlining the technical strategies for ensuring seamless interoperability through Foreign Function Interfaces (FFIs). This blueprint will serve as the structural guideline for all new development, particularly for the hybrid C++/Rust components in File 149 and the Lua-based scripting for hardware interaction.

The language roles are clearly delineated based on their respective strengths and suitability for specific tasks within the Aletheion ecosystem:

- **Rust:** Assigned to all core logic, business rules, and high-level application orchestration. Its unparalleled features for memory safety without garbage collection, fearless concurrency, and strong type system make it the ideal choice for building reliable and performant services. This includes the central modules in Files 146-150, such as `EmergencyEvacuationProtocol` and `ZeroKnowledgePassengerVerification`, as well as the overarching system architecture. Using `#![no_std]` in Rust is a key consideration for embedded applications, ensuring minimal runtime overhead [30].
- **C++:** Designated for all embedded sensor integration and performance-critical low-level operations. C++ is the industry standard for interacting directly with hardware, making it the correct choice for the sensor drivers and control logic in `transit_infrastructure.rs` (File 149), such as those managing cool pavement monitoring or misting stations . Its ability to provide fine-grained control over system resources is critical for real-time data acquisition and actuation.
- **Lua:** Chosen for sensor calibration, real-hardware engine scripting, and dynamic configuration. Lua is renowned for being lightweight, embeddable, and easy to

script, which makes it perfect for allowing operators and developers to modify system behavior at runtime without needing to recompile the entire application [39]. This is particularly valuable for calibrating sensors or scripting complex behaviors for autonomous vehicles and drones, providing a flexible interface between the static compiled code and the dynamic physical world.

Ensuring these distinct language components can work together harmoniously requires a robust FFI strategy. The provided context offers concrete technical solutions for bridging the communication gap between these languages.

For **Rust-to-C/C++ interoperability**, the standard practice involves exposing Rust functions through a C-compatible API. This can be achieved using the `cbindgen` tool, which automatically generates the necessary C header files from Rust code, simplifying the process of linking Rust libraries into a C or C++ project [32]. To manage the build process, modern build systems like CMake can be configured to invoke the Rust compiler (`cargo build`) as part of the overall compilation flow. This can be done using CMake's `add_custom_target` to create a dependency that ensures the Rust library is always up-to-date before the C++ code is compiled [15]. A more advanced and reliable solution is the `corrosion` project, which streamlines the process of calling Cargo from CMake, handling complexities like finding the correct Rust toolchain and managing build artifacts [15]. This approach allows a C++ application to link against a Rust-generated static or dynamic library and call its exported functions as if they were native C functions.

For **Lua-to-C/C++/Rust interoperability**, LuaJIT's Foreign Function Interface (FFI) extension presents a powerful and efficient option. LuaJIT's FFI allows Lua scripts to directly bind to and call C functions exposed in a shared library, eliminating the need for writing wrapper code in most cases [31] [33]. This is highly suitable for integrating C++ sensor drivers. A C++ driver can be compiled into a shared library exporting a simple C API. A Lua script running within Aletheion can then use the LuaJIT FFI to load this library and call the sensor functions directly, passing data back and forth with minimal overhead. While less common, it is also possible to create Lua bindings for Rust code. This would typically involve creating a thin C-compatible wrapper around the desired Rust functions, generating a Lua binding library using a tool like `tolua++`, and then loading this binding library in the Lua environment. This would enable high-level Lua scripts to orchestrate complex operations defined in performant Rust modules.

The following table summarizes the proposed language roles and the corresponding FFI strategies for key files, providing a clear architectural reference for developers.

File Number(s)	Path/File	Primary Language(s)	Role Definition	Key Interoperability Strategy
149	`transit_infrastructure.rs`	Rust + C++	Rust manages high-level state and logic; C++ handles direct, low-level interactions with environmental sensors (e.g., albedo readers, moisture sensors).	**Rust ↔ C++:** Use `cbindgen` to generate C headers from C++ sensor drivers. Call `cargo build` via CMake to integrate Rust and C++ builds. Rust consumes sensor data via the generated C API. [15] [32]
(Future)	Sensor Calibration Scripts	Lua	Provides a scripting interface for engineers to dynamically adjust sensor parameters (e.g., gain, offset) and test responses without recompiling firmware.	**Lua ↔ C++:** Use LuaJIT FFI to bind directly to the C API of the compiled C++ sensor drivers, enabling Lua scripts to read/write sensor calibration values. [31] [33]
(Future)	Real-Hardware Engine	Lua	Scripting the entire engine for real-hardware deployment, managing complex state machines for AVs, drones, and robotic arms.	**Lua ↔ Rust/C++:** High-level Lua scripts orchestrate sequences of calls to both Rust modules (for core logic) and C++ modules (for hardware control), acting as the mainline execution script. [39]

By adhering to this architectural blueprint, Aletheion can leverage the best tool for each job while maintaining a cohesive and integrated system. The explicit definition of language roles and the adoption of proven FFI techniques will prevent the creation of ad-hoc, brittle integrations and ensure the long-term health and extensibility of the codebase.

Automation-Driven Development with Documented Filler-Outputs

A significant departure from traditional software development practices is the proposed methodology for handling incomplete logic during Aletheion's next phase. Instead of generic, empty placeholder files that hinder progress and obscure intent, the project will adopt an automation-driven workflow centered on intelligent, self-documenting "filler-outputs." This innovative approach transforms the period of waiting for research results or final specifications from a manual, error-prone bottleneck into an automated, traceable, and productive state. This section details the conceptual framework, implementation strategy, and benefits of this workflow, ensuring that every generated file, regardless of its completeness, contributes positively to build integrity and progress tracking.

The user's directive specifies that filler-outputs should replace traditional placeholders. These outputs must be more than just skeleton code; they must be designed to actively guide the development process. An intelligent filler-output has several key characteristics. First, it contains the full structural boilerplate, including function signatures, comments, and necessary imports, ensuring that the file compiles and integrates seamlessly into the build system. Second, and most importantly, it is rich with documentation that explicitly states what is missing, why it is missing, and what is required to complete it. This documentation acts as a permanent record of the project's dependencies and outstanding tasks. Third, it incorporates specific markers or metadata that can be parsed by automation scripts. These markers signal to downstream processes that the file contains incomplete logic and are linked to a central manifest that tracks the status of all such items.

The implementation of this workflow relies on a combination of templating engines and manifest-based tracking. A simple templating language, such as Go templates or Handlebars.js, can be used to generate the initial file structure. The template would contain predefined sections for the filler content, populated with the necessary documentation and marker syntax. When a developer needs to create a new file whose logic is not yet available, a master automation script invokes this template, populating the placeholders with the current date, the target file's purpose, and the known prerequisites. This ensures consistency and prevents human error in setting up these temporary files.

A standardized marker syntax is crucial for the automation to function. This syntax should be easily parsable and semantically meaningful. For example, a filler output in Rust might look as follows:

```
// ==========================================================================
// FILE: src/agriculture/crop_planning.rs
// STATUS: INCOMPLETE (PENDING RESEARCH)
// DATE: 2026-03-12
// ==========================================================================
// ALETHEION-FILLER-START: CropSelectionEngine
// This module is intentionally left blank pending final research data.
//
// REQUIRED DATA: USDA Web Soil Survey data for Maricopa County pH, salini
// COMPLIANCE TARGETS: E-Compliance (Water Conservation), TM-Compliance (E
// DEPENDENCY: File 151-160: Soil & Water Analysis.
// REFERENCE MANIFEST: manifests/tier2_manifest.yml
// ==========================================================================
```

```
/// The CropSelectionEngine is responsible for recommending optimal crop v
/// based on local environmental conditions and market demand.
///
/// TODO: Implement logic to integrate with finalized soil composition dat
/// TODO: Connect to the `EnvironmentalModel` to assess water availability
pub fn select_crops() -> Vec<String> {
    panic!("Logic not yet implemented. See above comments for details and
}
// ALETHEION-FILLER-END: CropSelectionEngine
```

This single block of code serves multiple purposes. It is a valid Rust module that will compile. It documents the exact nature of the incompleteness. It links the issue to a specific external data source and compliance goal. Most critically, it contains a unique identifier (`CropSelectionEngine`) that can be used by a central manifest file. This manifest, likely a YAML file, would track all filler-outputs across the codebase, recording their status, the last time they were checked, and the progress of their dependent research tasks.

The manifest-based workflow creates a living document of the project's progress. A continuous integration (CI) pipeline or a dedicated maintenance script can periodically parse this manifest. If a research team completes a required data task, they update the manifest accordingly. The CI script, detecting this change, can then automatically identify all related filler-outputs, regenerate them with updated instructions, and trigger a rebuild of the dependent modules. This closes the loop, transforming the filler-output from a passive placeholder into an active component of the build and dependency-tracking system. This method ensures that no piece of incomplete logic is ever forgotten and that the transition from "incomplete" to "complete" is a transparent and automated process.

The benefits of this approach are substantial. It preserves build integrity at all times, preventing breakages caused by incomplete or structurally incorrect files. It provides a clear, auditable trail of what is known, what is unknown, and what is needed to move forward. It guides downstream developers and researchers by explicitly stating dependencies and requirements. Finally, it automates the management of the "waiting" state, freeing up developer time for productive work while ensuring that the project does not stall due to external dependencies. This workflow is a key enabler for the uninterrupted progression of Aletheion through Files 151-200 and beyond.

Integrated Execution Plan for Phased Implementation

To achieve the research goal of ensuring continuity and compliance for Aletheion's progression through Files 146–200, a structured, phased execution plan is required. This plan synthesizes the strategic pillars of ecological data sovereignty, a strict multi-language architecture, and an automation-driven development workflow. It provides a clear roadmap, detailing the immediate actions, concurrent tasks, and future implementation steps necessary to navigate the project from its current state (File 145 complete) to the successful generation of all files in the specified range. The plan is designed to eliminate bottlenecks, mitigate risk, and ensure that every line of code is grounded in verified data and aligned with Aletheion's core principles of ecological accuracy, treaty compliance, and architectural integrity.

The execution plan is divided into three distinct phases, each with specific objectives and deliverables. This phased approach allows for focused effort, parallel task execution, and systematic validation before proceeding to the next stage.

Phase 1: Immediate Action - Transit Bridge Completion (Files 146-150) This phase is the immediate priority and carries no research blockers, allowing for rapid and uninterrupted development .

- **Objective:** Complete the Tier 1 "Transit Bridge" by generating all five files in the `aletheion-mob/transit` module.
- **Actions:**

```
1.  Begin with `File 146: emergency_evacuation.rs`. Adhere strictly to the
2.  Proceed sequentially through `File 147: transit_security.rs`, focusing
3.  Generate `File 148: multi_modal_integration.rs`, creating a unified ro
4.  Develop `File 149: transit_infrastructure.rs`, implementing the dual-l
5.  Conclude with `File 150: transit_compliance.rs`, ensuring the `Indigen
```

- **Deliverables:** Five fully functional, documented, and compliant Rust modules. During this phase, the development of the automation scripts and template files for the "filler-outputs" should be initiated.

Phase 2: Concurrent Research and Foundational Modeling (Post-File 150) Upon completion of the Transit Bridge, code generation for the next tier must pause to await

critical research findings . This phase focuses on resolving the data sovereignty gaps while concurrently developing foundational models.

- **Objective:** Secure all necessary domain-specific data for Tier 2 (Agriculture & Environmental) and develop a proof-of-concept for cyber-aquatic system modeling.
- **Actions:**

```
1.  <strong>Initiate Parallel Research:</strong> Assign tasks to the desig
2.  <strong>Begin Foundational Modeling:</strong> Start the internal model
3.  <strong>Develop Automation Infrastructure:</strong> Finalize the desig
```

- **Deliverables:** Verified datasets for all Tier 2 modules, a validated architectural concept for VM-node based ecological modeling, and a fully operational automation framework for managing incomplete code.

Phase 3: Implementation - Agriculture, Environment, and Cross-Layer Integration (Files 151-200) This phase covers the bulk of the requested file range. It begins only after the prerequisites of Phase 2 are met, ensuring that all development is based on solid, verified foundations.

- **Objective:** Systematically generate all remaining files (151-200) using the defined multi-language architecture and the automation-driven workflow.
- **Actions:**

```
1.  <strong>Tier 2 (Agriculture & Environmental - 151-175):</strong> For e
2.  <strong>Tier 3 (Cross-Layer Integration - 176-200):</strong> With the
3.  <strong>Maintain Automation:</strong> Throughout this phase, the autom
```

- **Deliverables:** A fully generated and integrated set of code for Files 151-200, with all modules either fully functional or properly documented as pending further work. The codebase will be compliant, maintainable, and built upon a transparent and auditable foundation of data and design decisions.

By executing this integrated plan, the Aletheion project can systematically overcome its current challenges and ensure a smooth, compliant, and ecologically accurate progression through its next major milestone. This structured approach mitigates risk, optimizes resource allocation, and establishes a robust development culture centered on data sovereignty, architectural discipline, and intelligent automation.

Reference

1. (PDF) Integrating generic sensor fusion algorithms with sound state ... https://www.academia.edu/3528550/Integrating_generic_sensor_fusion_algorithms_with_sound_state_representations_through_encapsulation_of_manifolds
2. Quantitative Representativeness and Constituency of the Long-Term ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10460301/
3. Applications of implementation science in integrated conservation + ... https://journals.plos.org/climate/article?id=10.1371/journal.pclm.0000268
4. bing.txt - FTP Directory Listing ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/bing.txt
5. Ashley's Backward Phonemes | PDF | Nature - Scribd https://www.scribd.com/doc/135904238/Ashley-sBackwardPhonemes
6. Published Password Lists: 1 - Ineapple https://ineapple.com/known_pass1
7. [PDF] Neuromorphic Perception for Greenhouse Technology Using Event ... https://core.ac.uk/download/544084456.pdf
8. [PDF] Southwestern Desert Resources - AWS https://usgs-cru-individual-data.s3.amazonaws.com/kboykin/intellcont/SWDesertResources_Thomasetal2010-1.pdf
9. (PDF) Habitat Relations of the Ivory-billed Woodpecker on the ... https://www.researchgate.net/publication/262289174_Habitat_Relations_of_the_Ivory-billed_Woodpecker_on_the_Singer_Tract_Louisiana_A_Retrospective_Analysis
10. SIGLAS Faatlist07 - 09 PDF | PDF | Emergency Management - Scribd https://es.scribd.com/document/350319381/SIGLAS-faatlist07-09-pdf
11. MONITORING ECOLOGICAL CONDITION IN THE WESTERN ... https://link.springer.com/content/pdf/10.1007/978-94-011-4343-1.pdf
12. (PDF) SEMANTIC PHYSICS: FROM PHENOMENOLOGICAL ... https://www.researchgate.net/publication/397543573_SEMANTIC_PHYSICS_FROM_PHENOMENOLOGICAL_FRAMEWORK_TO_FUNDAMENTAL_ONTOLOGY
13. The European Union – from the Platform for Economic Cooperation ... https://www.academia.edu/36575421/

The_European_Union_from_the_Platform_for_Economic_Cooperation_to_the_System_of_Human_Rights_Protection_

14. (PDF) Gravitas Semantica: A Theory of Held Tension - ResearchGate https://www.researchgate.net/publication/398123589_GRAVITAS_SEMANTICA_A_Theory_of_Held_Tension
15. Best way to call Cargo from CMake? - Stack Overflow https://stackoverflow.com/questions/62648369/best-way-to-call-cargo-from-cmake
16. Rust cargo.toml specify custom path for C linker and compiler https://stackoverflow.com/questions/62371337/rust-cargo-toml-specify-custom-path-for-c-linker-and-compiler
17. Aletheion: The Emergence of a Semantic Field -From Ignition to ... https://www.researchgate.net/publication/396033796_Aletheion_The_Emergence_of_a_Semantic_Field_-From_Ignition_to_Navigation
18. Evaluation metrics in medical imaging AI: fundamentals, pitfalls ... https://www.sciencedirect.com/science/article/pii/S3050577125000283
19. The METRIC-framework for assessing data quality for trustworthy AI ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11297942/
20. (PDF) AletheionV2: A Decoder-Only LLM with Intrinsic Epistemic ... https://www.researchgate.net/publication/401671810_AletheionV2_A_Decoder-Only_LLM_with_Intrinsic_Epistemic_System_on_a_5D_Riemannian_Manifold
21. Essential terminology and considerations for validation of non ... https://www.sciencedirect.com/science/article/pii/S2590157522003364
22. How To Solve Skynet | PDF | Uncertainty - Scribd https://www.scribd.com/document/957848518/How-to-Solve-Skynet
23. Simple Index https://mirrors.aliyun.com/pypi/simple/
24. How to structure an embedded application while decoupling ... https://stackoverflow.com/questions/61274672/how-to-structure-an-embedded-application-while-decoupling-application-and-driver
25. Benchmarking WebAssembly for Embedded Systems - ACM https://dl.acm.org/doi/10.1145/3736169
26. AI-Enhanced Embedded IoT System for Real-Time Industrial Sensor ... https://www.researchgate.net/publication/397690070_AI-Enhanced_Embedded_IoT_System_for_Real-Time_Industrial_Sensor_Calibration
27. Introduction to Embedded Rust Programming | PDF | Microcontroller https://www.scribd.com/document/713326853/The-Embedded-Rust-Book

28. Call for inputs - Indigenous Peoples Free, Prior and Informed ... https://www.ohchr.org/en/calls-for-input/2025/call-inputs-indigenous-peoples-free-prior-and-informed-consent-business-and
29. Free, prior, and informed consent, local officials, and changing ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11589003/
30. Experts for Eg-Digital-Library-Final-Wrapper - Linknovate https://www.linknovate.com/search/?query=eg-digital-library-final-wrapper
31. How to use LuaJIT's ffi module when embedding? - Stack Overflow https://stackoverflow.com/questions/5919743/how-to-use-luajits-ffi-module-when-embedding
32. Rust FFI and cbindgen: Integrating Embedded Rust Code in C https://dev.to/theembeddedrustacean/rust-ffi-and-cbindgen-integrating-embedded-rust-code-in-c-59f8
33. LuaJIT ffi Bindings - C++ - Epic Developer Community Forums https://forums.unrealengine.com/t/luajit-ffi-bindings/31777
34. Understand foreign function interface (FFI) and language binding https://stackoverflow.com/questions/5440968/understand-foreign-function-interface-ffi-and-language-binding
35. https://mirrors.aliyun.com/openwrt/releases/24.10.... https://mirrors.aliyun.com/openwrt/releases/24.10.0-rc3/packages/mipsel_24kc/packages/Packages
36. GIS有关软件、框架、代码、资源整理- XYSGIS - 博客园 https://www.cnblogs.com/xysgis/articles/16390017.html
37. Debian -- Prospective packages https://www.debian.org/devel/wnpp/prospective
38. ubuntupackages.cgi - Ultimate Debian Database https://udd.debian.org/cgi-bin/ubuntupackages.cgi
39. [XML] Planeta PythonBrasil https://planet.python.org.br/rss10.xml
40. https://mirrors.aliyun.com/openwrt/releases/22.03.... https://mirrors.aliyun.com/openwrt/releases/22.03.2/packages/arc_archs/packages/Packages
41. https://mirrors.aliyun.com/openwrt/releases/22.03.... https://mirrors.aliyun.com/openwrt/releases/22.03.2/packages/powerpc_8540/packages/Packages